

Choi echo: dynamical irreversibility and local decoherence in quantum many-body chaos

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Quantifying intrinsic irreversibility in open quantum dynamics is central to understanding decoherence and information loss in many-body systems. In this work, we introduce the Choi echo, which provides an operational interpretation of the purity of the Choi state, the state representation of a quantum channel, as a quantifier of the robustness of quantum correlations against local information erasure. We employ this framework to analyze the reduced dynamics of a subsystem and to test whether local decoherence probes quantum chaos in many-body systems. Across paradigmatic spin chain models, we show that while the Choi echo captures key dynamical features, it also exhibits intrinsic limitations that, in certain regions of parameter space, restrict its ability to resolve the integrable-to-chaos transition at the level of spectral correlations. In particular, we demonstrate that local decoherence can spuriously signal quantum chaos in integrable regimes, tracing them to the inability of a strictly local probe to distinguish efficient coherent transport from genuinely scrambling dynamics. Our results show that local decoherence signals are controlled by the entanglement generated between the probe and its environment during the dynamics, rather than by spectral correlations, clarifying the practical scope of local dynamical diagnostics.

I. INTRODUCTION

Open quantum systems generically exhibit decoherence and information loss due to their coupling to uncontrolled environments. Quantifying the intrinsic irreversibility of such dynamics is central to modern quantum physics, as it connects the microscopic description of subsystem–environment interactions to practical benchmarks in quantum information processing [1, 2]. In particular, characterizing the stability of reduced dynamics—for example, against perturbations or explicit information erasure—is essential for understanding phenomena ranging from the fidelity of quantum gates to the microscopic emergence of thermalization in many-body systems. For closed systems, the canonical tool to probe dynamical stability is the Loschmidt echo, which measures the recoverability of a quantum state under an imperfect time-reversal protocol [3, 4]. Recent experiments have implemented nested time-reversal sequences, such as higher-order out-of-time-order correlators (OTOCs), to diagnose ergodicity and scrambling through dynamical echoes [5]. However, extending this operational notion of dynamical reversibility to the rigorous framework of open systems remains challenging. It requires a formalism that explicitly incorporates the non-unitary character of quantum channels and the irreversible erasure of information induced by the environment. In this work, we address this challenge by focusing on the fundamental object governing open-system dynamics: the quantum channel itself. Rather than tracking the evolution of specific state trajectories, which conflate initial conditions

with dynamical properties, we exploit the isomorphism between quantum channels and quantum states [6, 7]. We introduce the *Choi echo* as a rigorous probe of dynamical reversibility. Defined as the purity of the channel’s Choi state, this metric provides an operational measure of the system’s ability to preserve correlations against local erasure. By isolating the structural properties of the dynamical map, the Choi echo serves as a generalized indicator of how effectively an environment acts as a sink for quantum information, regardless of the microscopic mechanism driving the loss.

This framework is particularly relevant for the study of quantum chaos in many-body systems. The onset of chaos is the microscopic mechanism that allows an isolated system to act as an efficient bath for its own subsystems, a process formalized by the Eigenstate Thermalization Hypothesis (ETH) [8, 9]. Traditionally, diagnosing this regime relies on the spectral statistics of the Hamiltonian (e.g., the mean level spacing ratio $\langle \tilde{r} \rangle$), which require computationally prohibitive access to the full energy spectrum [10, 11]. The challenges inherent to spectral diagnostics have motivated a shift toward local dynamical proxies, grounded in the observation that the decoherence dynamics of a single probe spin embedded in a spin chain can faithfully encode and reveal the spectral chaoticity of the entire many-body system [12, 13]. However, it remains an open question whether local irreversibility is a unique fingerprint of spectral chaos or if other dynamical regimes can mimic these signatures.

We apply the Choi echo to scrutinize this local decoherence and spectral chaos correspondence in three paradigmatic spin chain models: the mixed-field Ising model, the random-field Heisenberg model, and the XXZ model with a local defect. Our results demonstrate that while the Choi echo is a superior diagnostic to state-based purities—resolving decoupling transitions with higher pre-

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cision due to its sensitivity to unitality—it reveals a fundamental limitation of local probes. We report “false positives” for chaos in the integrable XXZ chain, where efficient coherent transport generates high local decoherence indistinguishable from chaotic scrambling. These findings elucidate that local probes primarily measure the strength of dynamical coupling and information propagation, challenging the assumption that local irreversibility, at least of a single spin, is a sufficient condition for identifying many-body quantum chaos.

This paper is organized as follows. Section II outlines the quantum channel formalism. Section III develops the operational interpretation of the Choi echo. Section IV details the methodology for benchmarking the echo against spectral statistics. Section V presents the physical models and the comparative numerical analysis. Finally, Section VI summarizes our findings and their implications for the characterization of open quantum dynamics.

II. MATHEMATICAL FRAMEWORK

A. Quantum channels

We describe the reduced dynamics of a quantum subsystem via the quantum channel formalism. Let $\mathcal{E}_t : \mathcal{B}(\mathcal{H}_S) \rightarrow \mathcal{B}(\mathcal{H}_S)$ denote the completely positive and trace-preserving (CPTP) map governing the evolution of the subsystem density matrix ρ_S at time t [1, 14].

We consider a closed quantum system partitioned into a subsystem S and an environment E . The global system evolves under a unitary operator U , which may arise either from continuous-time Hamiltonian dynamics or from a discrete sequence of gates. Assuming the total system is initialized in a product state $\rho_{SE}(0) = \rho_S \otimes \rho_E$, the reduced map $\mathcal{E}_t$ is obtained by tracing out the environment,

$$\rho_S(t) = \mathcal{E}_t(\rho_S) = \text{Tr}_E[U(\rho_S \otimes \rho_E)U^\dagger]. \quad (1)$$

Although the global evolution U is reversible, the reduced dynamics is generally nonunitary due to system–environment entanglement. Equation (1) provides a complete specification of the quantum channel $\mathcal{E}_t$ independently of the input state and constitutes the system–environment representation of $\mathcal{E}_t$.

The quantum channel $\mathcal{E}_t$ admits an equivalent representation as a quantum state via the Choi–Jamiołkowski isomorphism [6, 7]. This duality establishes a one-to-one correspondence between any CPTP map and a positive semi-definite operator acting on a doubled Hilbert space.

Let $\mathcal{H}_{S'} \simeq \mathcal{H}_S$ be an auxiliary space, and define the maximally entangled state $|\Phi^+\rangle = d_S^{-1/2} \sum_i |i\rangle_S \otimes |i\rangle_{S'}$, where $\{|i\rangle\}$ is an orthonormal basis of $\mathcal{H}_S$. The Choi state associated with $\mathcal{E}_t$ is

$$\mathcal{D}(t) = (\mathcal{E}_t \otimes \mathcal{I}_{S'}) (|\Phi^+\rangle\langle\Phi^+|), \quad (2)$$

where $\mathcal{I}_{S'}$ is the identity channel on S' [14]. The operator $\mathcal{D}(t)$ acts on $\mathcal{H}_S \otimes \mathcal{H}_{S'}$ and uniquely encodes all properties of $\mathcal{E}_t$. The resulting matrix $\mathcal{D}(t)$ is a density matrix—i.e. a positive semi-definite matrix of unit trace—acting on the composite space $\mathcal{H}_S \otimes \mathcal{H}_{S'}$.

The Choi state $\mathcal{D}(t)$ encodes fundamental properties of the corresponding quantum channel $\mathcal{E}_t$. Complete positivity and trace preservation of $\mathcal{E}_t$ translate into $\mathcal{D}(t) \geq 0$ and $\text{Tr}_{S'}[\mathcal{D}(t)] = \mathbb{I}_S$, respectively. The rank of $\mathcal{D}(t)$ equals the minimum environment dimension required for a *minimal dilation* of $\mathcal{E}_t$ —i.e., a system–environment representation of $\mathcal{E}_t$ [cf. Eq. (1)] with the smallest environment dimension possible. Moreover, $\mathcal{E}_t$ is entanglement-breaking—erasing all quantum correlations between the system and any external system—if and only if the Choi state is separable across $S:S'$ bipartition.

B. Purity of the Choi state

A key scalar associated with the Choi state is its purity,

$$\text{Tr}[\mathcal{D}(t)^2] = \sum_{k,l} \text{Tr}[\mathcal{E}_t(|k\rangle\langle l|) \mathcal{E}_t(|l\rangle\langle k|)], \quad (3)$$

which quantifies the coherence-preserving character of the channel. The second equality follows directly from the definition of the Choi state in Eq. (2) and expresses the purity in terms of the action of the channel on the operator basis $\{|k\rangle\langle l|\}$ associated with an orthonormal basis $\{|k\rangle\}$ of $\mathcal{H}_S$. It satisfies

$$\frac{1}{d_S^2} \leq \text{Tr}[\mathcal{D}(t)^2] \leq 1.$$

The upper bound is achieved if and only if the reduced dynamics is unitary, while the minimum corresponds to the completely depolarizing channel that maps any input state to the maximally mixed state $\mathbb{I}_S/d_S$.

This quantity distinguishes dynamical regimes that state-based metrics cannot. For example, a full amplitude-damping channel maps all pure inputs to a pure ground state, yielding an output-state purity $\text{Tr}[\rho_S(t)^2] = 1$ for any pure input, identical to that of a unitary process. However, their Choi purities differ sharply: $\text{Tr}[\mathcal{D}^2] = 1$ for a unitary channel and $1/2$ for full amplitude damping. The Choi purity therefore probes the process coherence rather than the purity of individual trajectories.

To clarify the relation between the Choi and output-state purities, we derive the Haar-averaged output purity over pure inputs $|\psi\rangle$. Using standard identities for Haar integration [15], we obtain

$$\mathbb{E}[\text{Tr}(\mathcal{E}_t(|\psi\rangle\langle\psi|)^2)] = \frac{1}{d_S(d_S+1)} \left(\text{Tr}[\mathcal{E}_t(\mathbb{I}_S)^2] + d_S^2 \text{Tr}[\mathcal{D}(t)^2] \right). \quad (4)$$

The term $\text{Tr}[\mathcal{E}_t(\mathbb{I}_S)^2]$ quantifies deviations from unitality, since $\text{Tr}[\mathcal{E}_t(\mathbb{I}_S)^2] \geq d_S$ with equality if and only if $\mathcal{E}_t$ is unital.

Equation (4) shows that the average state purity contains two independent contributions: the Choi purity $\text{Tr}[\mathcal{D}(t)^2]$, capturing the intrinsic coherence preservation of the channel, and the unitality term, reflecting population-transfer effects. For unital channels, $\text{Tr}[\mathcal{E}_t(\mathbb{I}_S)^2] = d_S$, making the averaged output purity directly proportional to the Choi purity. For non-unital processes such as amplitude damping, the unitality term grows substantially, compensating for the decay of $\text{Tr}[\mathcal{D}(t)^2]$ and thereby obscuring irreversibility when assessing only output-state purity. Thus, Eq. (4) clarifies why state purity alone fails to detect structural information loss in nonunital dynamics.

III. THE CHOI ECHO: OPERATIONAL INTERPRETATION OF THE CHOI PURITY

We now establish an operational interpretation of the Choi purity [cf. Eq. (3)] by showing that it plays the role of an echo quantifying recoverability under local loss of information. In this formulation, the Choi purity is not merely a static functional of the channel, but a probe of dynamical reversibility analogous in structure to an echo protocol. This perspective highlights its ability to diagnose how strongly the global evolution becomes irreversibly correlated with degrees of freedom that are inaccessible to the subsystem.

The Loschmidt echo is a fundamental tool for characterizing the stability of quantum evolution, widely employed in contexts ranging from decoherence theory to critical many-body phenomena [3, 4]. Defined as $L(t) = |\langle \psi_0 | e^{i(H+\Sigma)t} e^{-iHt} | \psi_0 \rangle|^2$, it quantifies the ability of a system to recover its initial state after a forward evolution H , a perturbation Σ , and a backward evolution. Its decay reflects the system's sensitivity to perturbations and serves as a canonical measure of dynamical irreversibility.

In contrast to perturbations of the Hamiltonian, the Choi purity $\text{Tr}[\mathcal{D}(t)^2]$ acts as an echo that probes the stability of the global evolution against a *local* loss of information. Consider the channel $\mathcal{E}_t$ induced by the joint unitary $U(t)$ acting on the initially uncorrelated state $\rho_S \otimes \rho_E$, with $\rho_E = |\psi\rangle\langle\psi|$ any pure state. Combining the Choi definition in Eq. (2) with the system–environment representation of the channel in Eq. (1), the purity of the Choi state can be recast in the form

$$\text{Tr}(\mathcal{D}^2) = \left\langle \psi \left| \text{Tr}_S \left[U^\dagger \Lambda_S \left[U \left(\frac{\mathbb{I}_S}{d_S} \otimes |\psi\rangle\langle\psi| \right) U^\dagger \right] U \right] \right| \psi \right\rangle. \quad (5)$$

Here, $\Lambda_S(\cdot) = (\mathbb{I}_S/d_S) \text{Tr}_S(\cdot)$ represents the completely depolarizing channel acting locally on the subsystem S . For compactness, we have suppressed the explicit time arguments in this expression, denoting $U \equiv U(t)$ and

$\mathcal{D} \equiv \mathcal{D}(t)$.

Equation (5) shows that $\text{Tr}[\mathcal{D}(t)^2]$ is the fidelity with which the environment returns to its initial state after a three-step protocol:

1. A forward evolution under U , starting from a product state in which S carries no information,
2. A local “perturbation” is applied via a completely depolarizing channel Λ_S , on the system S only, that breaks all entanglement between S and E acquired during the forward evolution.
3. A backward evolution under $U^\dagger$.

The Choi purity therefore quantifies the recoverability of the environment following the loss of subsystem information and plays the role of an *echo*. In consequence, we refer to this quantity as the *Choi echo*.

Since the Choi state provides a basis-independent representation of the channel, the Choi echo quantifies reversibility at the level of the reduced dynamics. A high echo fidelity signifies that the forward evolution has generated only weak non-local correlations, so that the global state remains recoverable even after the subsystem has been depolarized. Conversely, a decaying echo signals dynamical irreversibility: correlations between S and E have become sufficiently intricate that severing them at the subsystem level eliminates the possibility of reconstructing the global state. This operational viewpoint identifies the Choi echo as a tool for diagnosing irreversibility and information scrambling in many-body systems. In parallel to entropy-based indicators, which quantify locally inaccessible information, and out-of-time-ordered correlators (OTOCs), which probe operator growth, the Choi echo provides a direct measure of the recoverability of global dynamics in the presence of local information loss.

IV. THE CHOI ECHO AS A PROBE FOR QUANTUM CHAOS

The characterization of quantum chaos in many-body systems is central to understanding thermalization, information scrambling, and the emergence of statistical mechanics from unitary dynamics [9]. Quantum-chaotic behavior manifests itself through several universal features: (i) spectral statistics governed by Random Matrix Theory (RMT) [10], (ii) eigenstate properties consistent with the eigenstate thermalization hypothesis (ETH) [8], and (iii) extreme sensitivity to perturbations, captured by Loschmidt echoes and out-of-time-ordered correlators (OTOCs). These signatures reflect distinct facets of the same phenomenon—namely, the progressive spreading of quantum information over the many-body Hilbert space. In practice, however, diagnosing chaos is challenging: spectral probes require access to the dense interior of the spectrum, while OTOCs and echo protocols demand controlled perturbations and precise reversals [16–19]. This

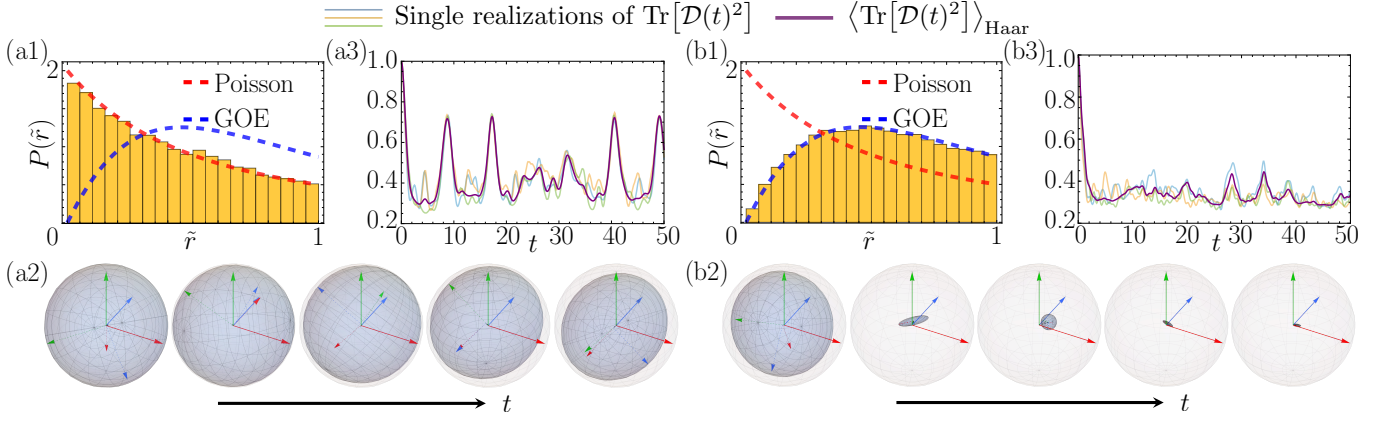


Figure 1. Spectral and single-spin dynamical behavior in regular and chaotic regimes of the mixed-field Ising model [Eq. (9)]. Panel (a) corresponds to an integrable regime $[(h_z, J) = (1.446, 0.05)]$ and the panel (b) to a chaotic regime $[(h_z, J) = (0.48, 0.8)]$, with fixed $h_x = 1$. **(a1)**, **(b1)** Distribution of nearest-neighbor level spacing ratios $P(\tilde{r})$ for $L = 16$ (even sector), compared to Poisson (red dashed) and GOE (blue dashed) predictions. **(a2)**, **(b2)** Action of the quantum channel $\mathcal{E}_t$ on the Bloch sphere of the probe spin ($L = 7$). The chaotic channel induces a rapid contraction of the sphere volume, contrasting with the coherent precession in the regular case. **(a3)**, **(b3)** Time evolution of the Choi echo $\text{Tr}[\mathcal{D}(t)^2]$ ($L = 7$). Low-opacity lines represent single realizations with random initial product states of the environment; the thick purple line corresponds to the analytical Haar-averaged Choi echo $\mathbb{E}[\text{Tr}[\mathcal{D}^2(t)]]$ derived in Eq. (7).

motivates the development of diagnostics that are more computationally tractable and physically accessible.

Focusing first on spectral indicators, chaotic systems display level repulsion consistent with RMT, whereas integrable models exhibit the uncorrelated spectrum characteristic of Poisson statistics [20]. A robust measure of these short-range correlations is the mean level spacing ratio $\langle \tilde{r} \rangle$, defined as the average of $\tilde{r}_n = \min(s_n, s_{n-1}) / \max(s_n, s_{n-1})$ for consecutive spacings $s_n = E_{n+1} - E_n$, evaluated within a fixed symmetry sector [21]. As illustrated in Figs. 1(a1) and (b1) for the mixed-field Ising chain [cf. Eq. (9)], the empirical distribution shifts from the Poissonian form, $P_P(\tilde{r}) = 2/(1+\tilde{r})^2$, to the Wigner-Dyson surmise characteristic of the Gaussian Orthogonal Ensemble (GOE), $P_{\text{GOE}}(\tilde{r}) = 8(\tilde{r} + \tilde{r}^2)/[7(1 + \tilde{r} + \tilde{r}^2)^{5/2}]$, marking the transition from integrable to chaotic behavior [22]. Although higher-order ratios mitigate the need for explicit desymmetrization [23], all spectral probes fundamentally require resolving large portions of the many-body spectrum, a task that becomes prohibitive for large Hilbert spaces [11, 24–26].

These limitations have motivated the search for accessible dynamical probes. Under chaotic dynamics, canonical typicality implies that small subsystems equilibrate toward stationary states that depend only on global conserved quantities [27, 28]. In the context of spin chains, the degree of equilibration of a single probe spin—typically identified with the first site of the chain—has been shown to correlate strongly with spectral signatures of chaos of the entire many-body system [13]. Quantita-

tively, this is captured by the averaged subsystem purity,

$$\bar{\mathcal{P}} = \frac{1}{N} \sum_{i=1}^N \left(\frac{1}{T} \int_0^T \text{Tr}[\rho_{S,i}^2(t)] dt \right), \quad (6)$$

computed over N random initial product states and a time window $[0, T]$. Across diverse spin models, $\bar{\mathcal{P}}$ exhibits a strong anticorrelation with $\langle \tilde{r} \rangle$, suggesting that local decoherence can serve as an experimentally friendly probe of the integrability-to-chaos transition.

However, a fundamental question remains: does local decoherence provide an unambiguous signature of many-body chaos, or can integrable dynamics produce similar behavior through mechanisms such as coherent transport? To address this, we employ the Choi echo framework introduced in Sec. III, applied to one-dimensional spin-1/2 chains of length L . We partition the system such that the first spin acts as the probe S , while the remaining $L - 1$ spins form the environment E . The probe's reduced dynamics are encoded in the quantum channel $\mathcal{E}_t$ generated by the global unitary $U(t) = e^{-iHt}$ [cf. Eq. (1)]. As shown in Figs. 1(a2) and (b2), integrable evolution preserves a coherent Bloch-sphere trajectory, whereas chaotic evolution induces a rapid contraction of the Bloch volume, signaling strong decoherence. Unlike state purity, which depends on the specific initial condition, the Choi echo $\text{Tr}[\mathcal{D}(t)^2]$ probes the intrinsic reversibility of the channel itself. By quantifying the stability of global correlations against a depolarizing operation on the probe, it provides a stricter and state-independent diagnostic of decoherence.

To isolate the generic dynamical properties of the Hamiltonian H from specific initial states of the environment, we compute the analytical Haar average of the

Choi echo over product states of the environment. Each environmental spin state is drawn independently from the Haar measure, a procedure corresponding to an infinite-temperature average [29]. As derived in Appendix A, this yields:

$$\mathbb{E}[\text{Tr}(\mathcal{D}^2)] = \sum_{\vec{\alpha}} \frac{4^{-L}}{3^{w(\vec{\alpha})}} \text{Tr} \left[\text{Tr}_S^2 [U(\sigma_1^0 \sigma_2^{\alpha_2} \dots \sigma_L^{\alpha_L}) U^\dagger] \right], \quad (7)$$

where $\vec{\alpha} \in \{0, x, y, z\}^{L-1}$, and for compactness we have suppressed the explicit time dependence. Here σ_j^μ denotes the Pauli operator labeled by $\mu \in \{0, x, y, z\}$ acting on site j (with $\sigma^0 \equiv \mathbb{I}$), and $w(\vec{\alpha})$ counts the number of non-identity operators in the environment string $\vec{\alpha} = (\alpha_2, \dots, \alpha_L)$. This analytical average is depicted in Figs. 1(a3) and (b3), where the Haar-averaged Choi echo (thick line) effectively filters out the fluctuations inherent to single environment realizations (low-opacity lines). To construct a direct analogue of $\langle \tilde{r} \rangle$ and $\overline{\mathcal{P}}$, we also average in time the Haar-averaged Choi echo:

$$\langle \text{Tr}[\mathcal{D}(t)^2] \rangle_{\text{Haar},t} = \frac{1}{T} \int_0^T \mathbb{E}[\text{Tr}[\mathcal{D}^2(t)]] dt. \quad (8)$$

The next section compares $\langle \tilde{r} \rangle$, $\overline{\mathcal{P}}$, and the Haar-averaged Choi echo across different models and parameter regimes.

V. MODEL SYSTEMS AND NUMERICAL ANALYSIS

We now assess the performance of the Choi echo as a dynamical probe for the transition to quantum chaos. To this end, we analyze three paradigmatic one-dimensional spin-1/2 models that exemplify distinct mechanisms of integrability breaking—competing fields, quenched disorder, and local defects. This diversity provides a stringent test of whether the echo can meaningfully separate chaotic from non-chaotic dynamics using only local information.

A. Spin Chain Hamiltonians

1. Mixed-field Ising model

We first consider the mixed-field Ising model with open boundary conditions, described by the Hamiltonian:

$$H = \sum_{i=1}^L (h_x \sigma_i^x + h_z \sigma_i^z) - J \sum_{i=1}^{L-1} \sigma_i^z \sigma_{i+1}^z, \quad (9)$$

where σ_i^α ($\alpha \in \{x, y, z\}$) are Pauli operators, L is the chain length, J is the nearest-neighbor Ising interaction, and h_x, h_z are transverse and longitudinal field strengths. This model possesses a global spatial reflection symmetry

$\mathcal{R} (i \leftrightarrow L-i+1)$, decomposing the Hilbert space into even and odd parity sectors. The model is integrable in two limits: the transverse-field Ising model ($h_z = 0$), solvable via Jordan-Wigner transformation [30], and the so-called classical Ising model ($h_x = 0$) [31]. When $J = 0$, the system decouples into a non-interacting set of L spins. The simultaneous presence of all three terms breaks integrability, rendering it a standard testbed for quantum chaos [32–34]. Notably, it has been rigorously proven that for any non-zero J , h_x , and h_z , the model possesses no non-trivial local conserved quantities [35]. The chaotic regime is frequently studied using parameters where all three terms are of comparable magnitude, $J \sim h_x \sim h_z$, as this maximizes the non-commutation responsible for driving chaotic behavior [36–38].

2. Heisenberg model with random fields

Next, we examine the Heisenberg model with random fields:

$$H = \frac{1}{4} \sum_{i=1}^{L-1} (\vec{\sigma}_i \cdot \vec{\sigma}_{i+1}) + \frac{1}{2} \sum_{i=1}^L h_i \sigma_i^z, \quad (10)$$

where h_i are independent random variables drawn uniformly from the interval $[-h, h]$, and h denotes the disorder strength. This model conserves the total magnetization $M_z = \sum_i \sigma_i^z$. This $U(1)$ symmetry is equivalent to conserving the total number of up spins, $N_\uparrow$, related to the magnetization via $\langle M_z \rangle = 2N_\uparrow - L$. The clean limit ($h = 0$) is integrable by Bethe-ansatz [39]. For $h > 0$, integrability is broken. This Hamiltonian describes the transition from an ergodic or thermal phase ($h \lesssim 3.5$) to a many-body localized (MBL) phase at strong disorder [21, 40, 41].

3. XXZ model with a local defect

Finally, we study the XXZ model with a single local defect:

$$H = \frac{1}{4} \sum_{i=1}^{L-1} [J_{xy} (\sigma_i^x \sigma_{i+1}^x + \sigma_i^y \sigma_{i+1}^y) + J_z \sigma_i^z \sigma_{i+1}^z] + \frac{1}{2} \varepsilon \sigma_d^z. \quad (11)$$

Here, J_{xy} and J_z are anisotropic interaction strengths, and ε is the field strength at a defect site d . Like the Heisenberg model, this system conserves M_z . The defect is placed near the center to break spatial reflection symmetry explicitly. For $\varepsilon = 0$, the model is integrable [39]. A non-zero defect $\varepsilon \neq 0$ breaks integrability [42–44], providing a minimal model where chaos is induced solely by a local perturbation on a single site. We note that in the limit $\varepsilon \gg 1$, the chain effectively decouples into two non-interacting parts.

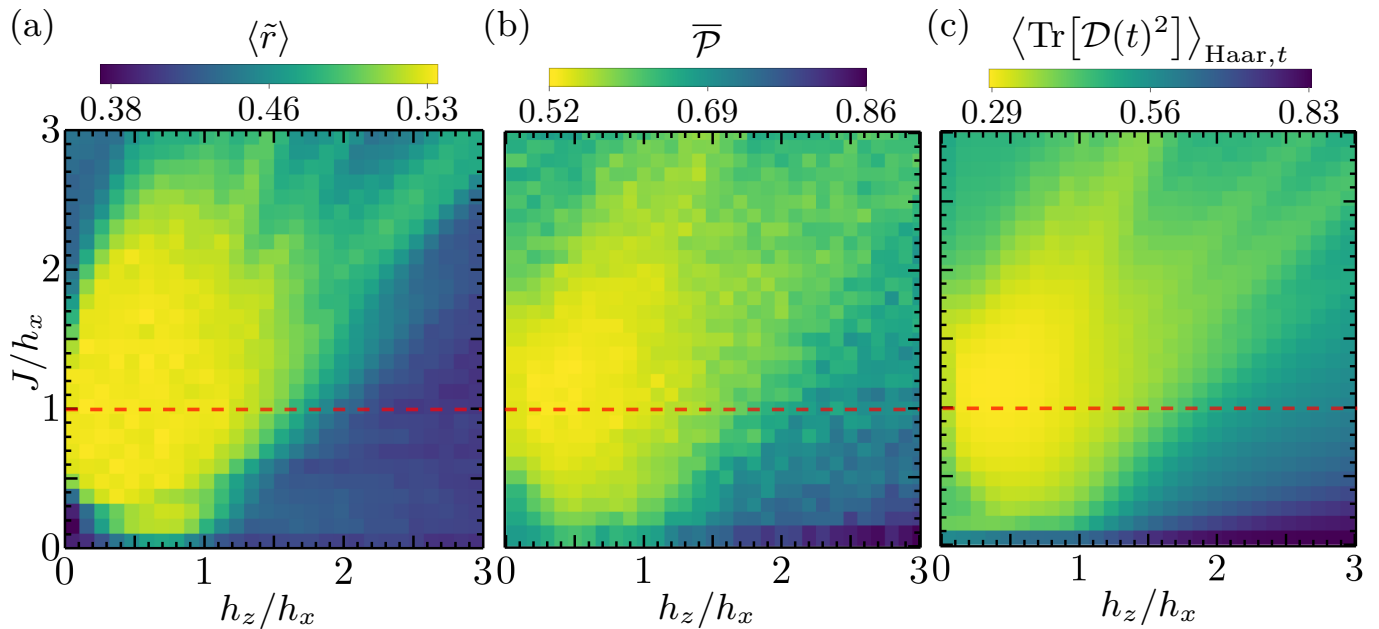


Figure 2. Comparison of short-range spectral statistics and local dynamical probes for the mixed-field Ising model [Eq. (9)] as a function of h_z/h_x and J/h_x . **(a)** Mean level spacing ratio $\langle \tilde{r} \rangle$, computed for $L = 16$ within the even symmetric sector. **(b)**, **(c)** Local dynamical probes computed for the spin at the first site ($L = 7$): **(b)** averaged subsystem state purity $\overline{\mathcal{P}}$ [Eq. (6)] ($N = 50$, $T = 100$); **(c)** averaged Choi echo $\langle \text{Tr}[\mathcal{D}(t)^2] \rangle_{\text{Haar},t}$ [Eq. (8)] ($T = 100$). The horizontal red dashed line ($J/h_x = 1$) indicates the parameter cut where the correspondence between $\langle \tilde{r} \rangle$ and $\overline{\mathcal{P}}$ was analyzed in Ref. [13] for $h_z/h_x \in [0, 2.5]$.

B. Results

We benchmark the Haar- and time-averaged Choi echo $\langle \text{Tr}[\mathcal{D}(t)^2] \rangle_{\text{Haar},t}$ [Eq. (8)] against two reference indicators: (i) the averaged subsystem state purity $\overline{\mathcal{P}}$ [Eq. (6)], and (ii) the mean level spacing ratio $\langle \tilde{r} \rangle$, which provides the spectral ground for quantum many-body chaos. While both purities are computed for small chains ($L = 7$) with the probe at site 1, the spectral statistics are evaluated via exact diagonalization on significantly larger systems ($L = 16$ or $L = 18$). This mismatch in system sizes imposes a stringent requirement: a reliable local dynamical probe must be robust to finite-size effects and must recover the chaotic transition inferred from the spectral benchmark.

1. Resolving chaos and decoupling in the mixed-field Ising model

Fig. 2 compares the three quantities across the $(J/h_x, h_z/h_x)$ plane. The mean level spacing ratio is computed for $L = 16$ within the even sector (dimension 32,898). As shown in Fig. 2(a), it exhibits a well-defined chaotic dome where $\langle \tilde{r} \rangle \approx 0.53$. Despite the reduced system size, both $\overline{\mathcal{P}}$ and $\langle \text{Tr}[\mathcal{D}(t)^2] \rangle_{\text{Haar},t}$ reproduce this structure, with low purity correlating with high $\langle \tilde{r} \rangle$. Notably, the Choi echo [Fig. 2(c)] resolves the boundaries of the chaotic region more sharply than the state purity [Fig. 2(b)].

However, a discrepancy emerges in the regular region $1.5 \leq h_z/h_x \leq 3$ under weak coupling $J/h_x < 0.5$, indicating that the dynamical probes do not anticorrelate with the spectral statistics across the full parameter space. In this regime, the mean level spacing ratio $\langle \tilde{r} \rangle$ saturates at its Poisson value (≈ 0.38), whereas both dynamical probes display a smooth gradient. As J/h_x decreases, the Choi echo and the state purity increase gradually, reaching their maxima only when $J \rightarrow 0$. Thus, while spectral statistics detect a uniform regular phase, local probes remain sensitive to the competition between the Ising coupling J and the local fields (h_x, h_z) . The Choi echo, in particular, captures the continuous drift toward locally unitary dynamics as interactions become negligible.

Finally, in regions where the interaction J is comparable to the local fields, local probes regain a faithful anticorrelation with $\langle \tilde{r} \rangle$, as observed along the parameter cut indicated by the red dashed line in Fig. 2 ($J/h_x = 1$). This is precisely the regime analyzed in Ref. [13] for $0 < h_z/h_x < 2.5$, where both dynamical and spectral indicators show consistent signatures of chaos.

2. Tracking the thermal-to-MBL transition in the Heisenberg model with random fields

In the Heisenberg model with random fields, we examine the transition from thermal to the MBL phase. Fig. 3 displays the spectral and dynamical metrics as a

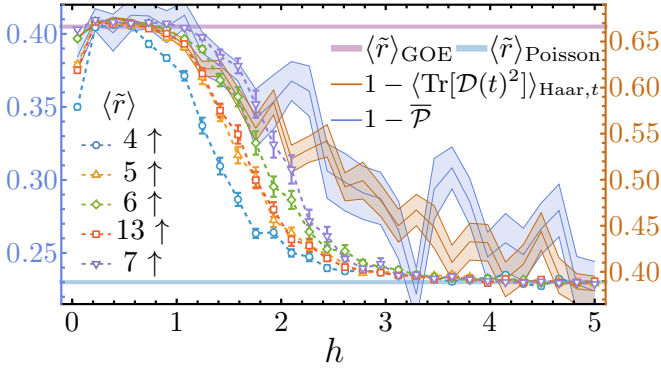


Figure 3. Comparison of spectral statistics and local dynamical probes for the Heisenberg model with random fields [Eq. (10)] as a function of disorder strength h . All quantities are averaged over 30 disorder realizations. **Symbols:** Mean level spacing ratio $\langle \tilde{r} \rangle$, computed for $L = 18$ within different magnetization sectors $N_\uparrow$ (horizontal lines indicate GOE and Poisson limits). **Solid lines:** Local dynamical probes computed for the spin at the first site ($L = 7$), plotted as deviations from unity (impurities): **Left axis (blue):** averaged subsystem state impurity $1 - \overline{\mathcal{P}}$ [Eq. (6)] ($N = 50$, $T = 100$); **Right axis (brown):** averaged Choi echo deviation $1 - \langle \text{Tr}[\mathcal{D}(t)^2] \rangle_{\text{Haar},t}$ [Eq. (8)] ($T = 100$). Shaded bands and error bars indicate the standard deviation.

function of disorder strength h , with all quantities averaged over 30 disorder realizations. The mean level spacing ratio $\langle \tilde{r} \rangle$ is computed for $L = 18$ across five magnetization sectors, specified by the number of up spins $N_\uparrow \in \{4, 5, 6, 7, 13\}$, with corresponding Hilbert space dimensions 3,060, 8,568, 18,564, 31,824, and 8,568, respectively. These results, shown as symbols with error bars, capture the crossover from GOE to Poisson statistics around $h_c \approx 3.5$. Here, we observe a robust agreement between the spectral statistics and the local probes. Both the deviation of the Choi echo from unity ($1 - \langle \text{Tr}[\mathcal{D}(t)^2] \rangle_{\text{Haar},t}$) and the state impurity ($1 - \overline{\mathcal{P}}$) remain high in the thermal phase ($h \lesssim 1$) and decay as the system transitions to the MBL phase. Significantly, in the deep thermal phase, the Choi echo exhibits negligible variance across disorder realizations (the standard deviation band in Fig. 3 is virtually indiscernible), suggesting it is a stable quantifier of the dynamical map's scrambling power. In contrast, the state purity shows larger fluctuations, reflecting its dependence on the specific initial state trajectory.

3. Transport-induced false positives for chaos in the XXZ model with a local defect

Finally, the results for the XXZ model with a local defect reveal a second fundamental limitation of local dynamical probes when contrasted with spectral statistics of the full chain. Fig. 4 shows intensity maps of $\langle \tilde{r} \rangle$, $\overline{\mathcal{P}}$, and $\langle \text{Tr}[\mathcal{D}(t)^2] \rangle_{\text{Haar},t}$ as functions of the anisotropy J_{xy}/J_z and defect strength ε/J_z . The spectral bench-

mark $\langle \tilde{r} \rangle$ [Fig. 4(a)] is computed for $L = 18$ in the $N_\uparrow = 7$ sector (dimension 31,824), with the defect placed at site $d = 9$. The dynamical probes [Figs. 4(b,c)] are obtained for $L = 7$ with the defect at site $d = 3$. In both cases, the defect lies in the bulk but away from the reflection axis, ensuring explicit parity breaking. This placement avoids accidental residual symmetries and makes the intensity maps qualitatively comparable despite the different system sizes.

The most prominent discrepancy appears in the region $J_{xy}/J_z > 1$ with a vanishing defect $\varepsilon \rightarrow 0$. Here the model approaches the clean XXZ chain and is integrable, a fact captured by $\langle \tilde{r} \rangle$, which saturates at the Poisson value. In sharp contrast, both dynamical probes yield low values that are indistinguishable from the chaotic regime. This constitutes a spurious signature of GOE statistics: strong interactions lead to fast local relaxation of the probe, overwhelming the global integrable structure and producing a false positive for chaos.

A second, more subtle discrepancy mirrors the behavior observed in the mixed-field Ising model. For $J_{xy}/J_z \lesssim 1$, the spectral indicator $\langle \tilde{r} \rangle$ exhibits a sharp transition toward regular behavior as J_{xy} decreases. The dynamical probes, however, respond through a smooth gradient: as $J_{xy} \rightarrow 0$, the flip-flop term responsible for spin transport vanishes and the probe dynamics crosses over continuously to an Ising-type dephasing channel. Thus, while spectral statistics detect the abrupt breakdown of GOE correlations, the local probes remain sensitive to the gradual suppression of transport rather than the transition to the integrable limit.

C. Discussion

The comparison across the three models yields a coherent picture of the diagnostic capabilities and limitations of local dynamical probes. The Choi echo quantifies the recoverability of global correlations after information on the probe is locally erased. Its magnitude reflects how effectively the environment can rebuild the correlations required for reversibility, rather than the spectral complexity encoded in the correlations of the spectrum.

In the Heisenberg model with random fields this recoverability is intrinsically suppressed. Chaotic evolution rapidly distributes information across the system and environment, and the depolarization on the probe removes the correlations needed to reconstruct the global state. As a result, both the Choi echo and the state purity track the integrability-to-chaos crossover with high fidelity, aligning with the behavior of short-range correlations of the spectrum.

However, the XXZ chain with a local defect exposes a key limitation: strong local relaxation does not uniquely signal spectral chaos. In this model, ballistic or diffusive transport generates rapid entanglement between the probe and the rest of the chain even in parameter regimes where the many-body spectrum remains

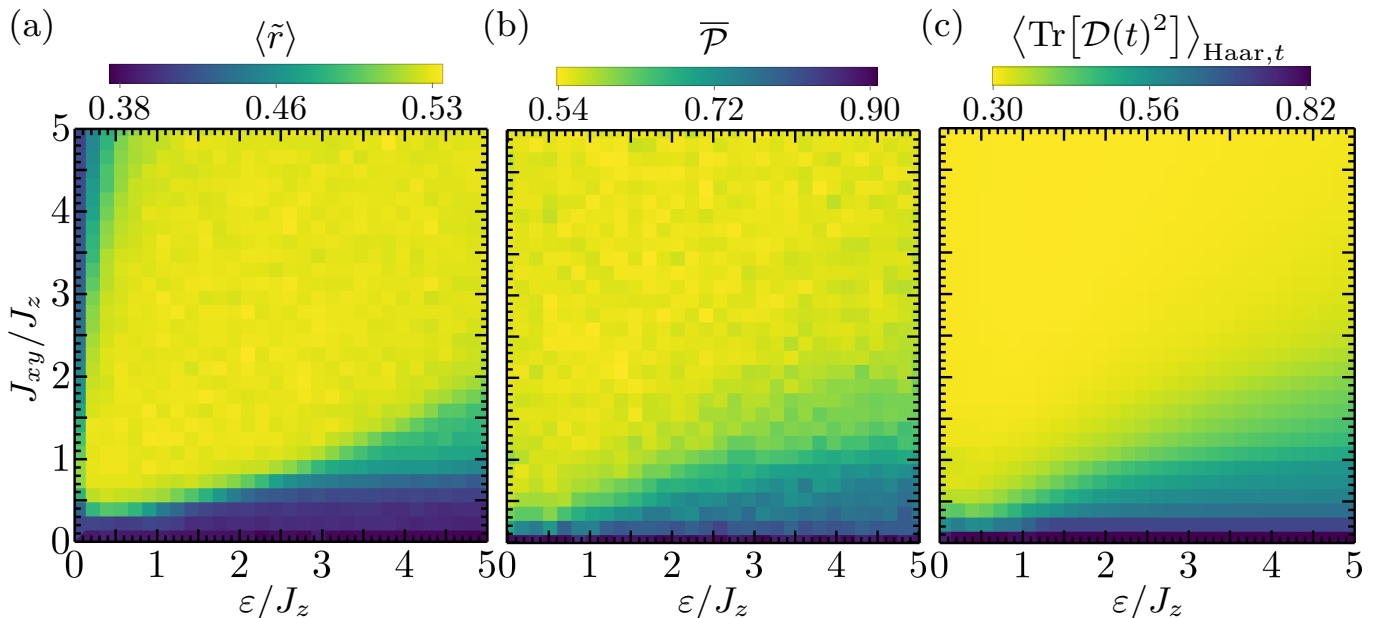


Figure 4. Comparison of spectral statistics and local dynamical probes for the XXZ model with a local defect [Eq. (11)] as a function of J_{xy}/J_z and ε/J_z . **(a)** Mean level spacing ratio $\langle \tilde{r} \rangle$, computed for $L = 18$ ($N_\uparrow = 7$) with the defect at site $d = 9$. **(b), (c)** Local dynamical probes computed for the spin at the first site ($L = 7$) with the defect at site $d = 3$: **(b)** averaged subsystem state purity $\overline{\mathcal{P}}$ [Eq. (6)] ($N = 50$, $T = 100$); **(c)** averaged Choi echo $\langle \text{Tr}[\mathcal{D}(t)^2] \rangle_{\text{Haar}, t}$ [Eq. (8)] ($T = 100$). In both configurations, the defect placement explicitly breaks spatial reflection symmetry.

integrable. From the standpoint of the Choi echo, these transport-driven processes are operationally indistinguishable from genuine scrambling—both mechanisms disperse correlations across the system, preventing their local reconstruction. Consequently, local probes fundamentally diagnose the strength of dynamical coupling and the efficiency of information propagation, rather than the presence of correlations in the spectrum.

The mixed-field Ising model highlights a complementary advantage of the Choi echo. Because it characterizes the unitarity of the reduced dynamics instead of the mixedness of particular trajectories, it resolves the approach to the decoupled limit more sharply than state-based metrics. This distinction underscores the conceptual separation between probes of dynamical reversibility and probes of state purity: while both reflect aspects of equilibration, only the former directly quantify the intrinsic stability of the reduced dynamics under local perturbations.

From a technical standpoint, the Choi echo offers a more intrinsic diagnostic of reduced dynamics than the subsystem state purity. The analytical relationship established in Eq. (4) clarifies why state purity can remain high in non-unitary regimes, whereas the echo directly quantifies deviations from unitarity. This distinction proved essential in the mixed-field Ising model, where the echo resolved the approach to the decoupled limit with significantly higher fidelity.

Crucially, however, our results challenge the universality of the “local decoherence–spectral chaos in the entire system correspondence” between subsystem dynam-

ics and spectral statistics. While chaotic dynamics (as in the random-field Heisenberg model) reliably induce high decoherence, we demonstrate that the converse does not hold. The emergence of “false positives” in the XXZ model with a local defect, specifically within a parameter regime exhibiting Poissonian statistics, shows that single-spin probes cannot distinguish between genuine scrambling induced by many-body chaos and entanglement generation driven by coherent transport. In both regimes, the mechanism of information loss—from the perspective of the local subsystem—is operationally identical: the environment acts as an effective sink for correlations, rendering the local dynamics irreversible despite the underlying integrability of the global spectrum. We anticipate that such false positives are not unique to the XXZ chain but will arise generically in integrable systems perturbed by local defects [45], where the conserved quantities of the integrable limit are nonlocal and are all broken by the presence of a single-site defect, as well as in integrable systems where the entangling power can mimic the effect of quantum many-body chaos [46].

The combined analysis across the mixed-field Ising, random-field Heisenberg, and defected XXZ models demonstrates both the strengths and the boundaries of local dynamical indicators. While the echo reliably tracks the chaotic transition in regimes where spectral complexity and dynamical irreversibility coincide, it also reveals scenarios in which rapid local relaxation arises from mechanisms unrelated to many-body chaos. In particular, coherent transport in integrable models can mimic the relaxation patterns typically associated with

scrambling, underscoring that local probes fundamentally quantify the efficacy of information propagation rather than the fine structure of the spectrum.

VI. CONCLUSIONS

We have discussed Choi-state purity as a physically motivated dynamical metric for characterizing reduced quantum evolution. By reformulating this quantity as the *Choi echo*, we made explicit that the diagnostic operates directly at the level of the dynamical map, independent of the behavior of particular input states—a distinction reflected in its analytical relation to the Haar-averaged state purity. This channel-based formulation shows that the unitarity of the reduced dynamics is governed by the extent to which the global evolution remains reversible once local information has been erased, thereby identifying reversibility as an intrinsic property of the map itself.

Our comparative analysis reveals that the Choi echo provides a more fundamental characterization of reduced dynamics than the subsystem state purity. We derived an analytical relationship [Eq. (4)] demonstrating that state purity is intertwined with the channel’s unitality. This theoretical insight explains our numerical findings in the mixed-field Ising model, where the Choi echo resolved the transition to the decoupling limit with significantly higher definition than state-based metrics. Consequently, for distinguishing between decoherence-free unitary evolution and genuine information loss, the Choi echo constitutes a superior diagnostic.

Our analysis shows that the degree of local decoherence is primarily a signature of the *strength of dynamical coupling* and the efficiency of information propagation, rather than a unique fingerprint of quantum chaos. While the Choi echo constitutes a rigorous metric for quantifying dynamical irreversibility—a property central to the study of thermalization and information scrambling—its use as a proxy for spectral chaos requires careful contextualization in terms of the system’s transport properties. Future work may address whether channel-based metrics involving multi-point correlations can bridge the gap between dynamical irreversibility and the fine-grained spec-

tral complexity of quantum chaos. We have thus established that the Choi echo is a rigorous and versatile probe of dynamical irreversibility in composite quantum systems. These findings motivate the exploration of channel-based probes involving multi-spin partitions or temporal correlation structures, which may bridge the gap between local irreversibility and spectral diagnostics of quantum chaos.

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Appendix A: Haar average of the Choi state purity $\text{Tr}[\mathcal{D}(t)^2]$

In this appendix, we derive the Haar-expectation value of the Choi state purity, Eq. (7). The channel $\mathcal{E}_t$ acts upon the first spin (S) in the chain, while the environment E consists of the $L - 1$ spins $i = 2, \dots, L$, prepared in a random product state $|\psi_E\rangle = \bigotimes_{i=2}^L |\phi_i\rangle$.

The Choi echo for this channel is related to the evolved environment state $\rho_E(t) = \text{Tr}_S[U(t)(\mathbb{I}_S/2 \otimes |\psi_E\rangle\langle\psi_E|)U^\dagger(t)]$ via $\text{Tr}[\mathcal{D}(t)^2] = d_S^2 \text{Tr}_E[\rho_E(t)^2]$ (where $d_S = 2$). We compute the Haar average of this quantity over $|\psi_E\rangle$ using the swap trick $\text{Tr}[A^2] = \text{Tr}[\mathbb{S}(A \otimes A)]$:

$$\mathbb{E}[\text{Tr}[\mathcal{D}(t)^2]] = \text{Tr}_{E_A E_B} [\mathbb{S}_E \mathbb{E}[\rho_E(t) \otimes \rho_E(t)]], \quad (\text{A1})$$

And then:

$$\mathbb{E}[\rho_E(t) \otimes \rho_E(t)] = \mathbb{E}\left[\text{Tr}_{S_A S_B} \left(U^{\otimes 2} \left(\frac{\mathbb{I}_S}{2} \otimes |\psi_E\rangle\langle\psi_E| \right)^{\otimes 2} U^{\dagger \otimes 2} \right)\right] = \text{Tr}_{S_A S_B} \left[U^{\otimes 2} \left(\frac{\mathbb{I}_S^{\otimes 2}}{4} \otimes \mathbb{E}[|\psi_E\rangle\langle\psi_E|^{\otimes 2}] \right) U^{\dagger \otimes 2} \right]. \quad (\text{A2})$$

Due to the product state structure, the environment av-

erage factorizes:

$$\mathbb{E}[|\psi_E\rangle\langle\psi_E|^{\otimes 2}] = \bigotimes_{i=2}^L \mathbb{E}[|\phi_i\rangle\langle\phi_i|^{\otimes 2}] = \bigotimes_{i=2}^L \left(\frac{\mathbb{I}_i + \mathbb{S}_i}{6} \right). \quad (\text{A3})$$

Here, $\mathbb{E}[|\phi\rangle\langle\phi|^{\otimes 2}] = (\mathbb{I} + \mathbb{S})/(d(d+1))$ is the standard second moment [47], which for qubits ($d = 2$) yields $(\mathbb{I} + \mathbb{S})/6$.

Substituting Eq. (A2) and Eq. (A3) into Eq. (A1):

$$\mathbb{E}[\text{Tr}[\mathcal{D}(t)^2]] = \frac{1}{4} \text{Tr}[(\mathbb{I}_S^{\otimes 2} \otimes \mathbb{S}_E)U^{\otimes 2} \times \left(\mathbb{I}_S^{\otimes 2} \otimes \bigotimes_{i=2}^L \frac{\mathbb{I}_i + \mathbb{S}_i}{6}\right)U^{\dagger \otimes 2}]. \quad (\text{A4})$$

We expand the averaged projector in the Pauli basis using $\frac{\mathbb{I} + \mathbb{S}}{6} = \frac{1}{12} \sum_{k \in \{0, x, y, z\}} c_k(\sigma_k \otimes \sigma_k)$, where $c_0 = 3$ and

$c_{x,y,z} = 1$. The tensor product becomes

$$\begin{aligned} \bigotimes_{i=2}^L \frac{\mathbb{I}_i + \mathbb{S}_i}{6} &= \frac{1}{12^{L-1}} \sum_{\vec{\alpha}} \left(\prod_{j=1}^{L-1} c_{\alpha_j} \right) (\sigma_{\vec{\alpha}}^E \otimes \sigma_{\vec{\alpha}}^E) \\ &= \frac{1}{12^{L-1}} \sum_{\vec{\alpha}} 3^{L-1-w(\vec{\alpha})} (\sigma_{\vec{\alpha}}^E \otimes \sigma_{\vec{\alpha}}^E), \end{aligned} \quad (\text{A5})$$

where $\vec{\alpha} = (\alpha_2, \dots, \alpha_L) \in \{0, x, y, z\}^{L-1}$, $\sigma_{\vec{\alpha}}^E = \bigotimes_{i=2}^L \sigma_{\alpha_i}$, and $w(\vec{\alpha})$ counts the non-identity operators in $\vec{\alpha}$.

Defining the full Pauli operator $\sigma_{\vec{\alpha}}^F = \mathbb{I}_1 \otimes \sigma_{\vec{\alpha}}^E$ (using $\sigma_0 \equiv \mathbb{I}$), and inserting the expansion, we find

$$\begin{aligned} \mathbb{E}[\text{Tr}[\mathcal{D}(t)^2]] &= \frac{1}{4} \cdot \frac{1}{12^{L-1}} \sum_{\vec{\alpha}} 3^{L-1-w(\vec{\alpha})} \text{Tr}[(\mathbb{I}_S^{\otimes 2} \otimes \mathbb{S}_E)(U(\sigma_{\vec{\alpha}}^F)U^\dagger \otimes U(\sigma_{\vec{\alpha}}^F)U^\dagger)] \\ &= \frac{1}{4^L} \sum_{\vec{\alpha} \in \{0, x, y, z\}^{L-1}} \frac{1}{3^{w(\vec{\alpha})}} \text{Tr}_E[(\text{Tr}_S[U(t)\sigma_{\vec{\alpha}}^F U^\dagger(t)])^2]. \end{aligned} \quad (\text{A6})$$

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